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Between Sustainability and Contradictions: Multi-Level Effects of Social Acceptance in Indonesian Geothermal Energy Projects

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Abstract: The Indonesian government promotes geothermal energy development to support national decarbonization goals and international climate commitments. However, several geothermal projects faced prolonged community resistance, including in Padarancang, where opposition persisted for more than fifteen years. This article examines social acceptance not as a community-level attitude alone, but as an outcome shaped by interactions across multiple governance levels. This study employed a qualitative case study approach. Data were collected through semi-structured interviews, observations, and document analysis, then analyzed thematically. The findings showed that community resistance was not driven by misinformation or limited awareness. Instead, it constituted a structured and reflective political response to exclusionary governance practices. Although the project enjoyed strong socio-political and market acceptance at the national level (supported by policy frameworks, investment instruments, and local government compliance), this legitimacy did not translate downward. A gap emerged between formal, policy-based legitimacy and social legitimacy at the community level. Low community acceptance was primarily driven by limited participation in decision-making. It also stemmed from perceived environmental, social, and cultural risks that were not balanced by meaningful local benefits. The study further demonstrated that multi-level governance (MLG) in geothermal development operated predominantly in a top-down and disciplinary manner. Authority was centralized, while responsibility for managing social conflict was displaced to local actors. This paper reconceptualizes social acceptance as a cross-level governance outcome. It shows how misalignment across governance scales can undermine renewable energy transitions and contribute to the failure of achieving national and international energy mix targets.

Keywords: Social acceptance; Local community; Multi-level governance; Renewable energy; Geothermal

1 Introduction

Indonesia possesses approximately 40% of the world's geothermal energy potential, distributed across 359 locations with an estimated 23,465.5 MWe, including both resources and reserves, as shown in Table 1 [1, 2]. With this potential, geothermal energy plays a crucial role in the decarbonization of Indonesia's energy sector, especially in the electricity generation [3], because it has low emissions compared to fossil fuels [4], is reliable and sustainable [5], and geothermal energy can be cost-competitive with other fossil fuels in the long term [6].

Through Presidential Regulation No. 22/2017 National Energy General Plan (Rencana Umum Energi Nasional, RUEN) and its commitment to the Paris Agreement, the Indonesian government targets a 23% share of new and renewable energy (NRE) by 2025, with geothermal contributing approximately 8% of installed capacity [3, 7]. However, achieving these targets is hindered by persistent social resistance in several designated Geothermal Working Areas (WKP). Examples include Mount Talang [8], Kaldera Danau [9, 10], Mount Ciremai [11], Dieng [12], Mount Lawu [13], Baturaden [14], Bedugul [15], Matalako [16], and Wae Sano [17].

This study focuses on the Kaldera Danau WKP in Banten Province, which has been designated a national strategic project (Proyek Strategis Nasional, PSN) due to its estimated 790 MWe capacity. The Padarincang geothermal plant (Pembangkit Listrik Tenaga Panas Bumi, PLTP), with a planned output of 110 MWe, represents a critical test case for understanding why centrally supported projects fail locally despite strong policy backing. Despite Banten's substantial renewable energy mix (over 6,300 MWe from various sources), geothermal remains a strategic pillar due to its baseload capability. However, the Padarincang project has faced unresolved local opposition for over 15 years, highlighting a critical misalignment between national strategic design and local social legitimacy.

Table 1. The potential for geothermal energy development in Indonesia

Region	Number of Points	Energy Potential (MWe)				
		Resources		Reserves		
		Speculative	Hypothesis	Perhaps	Alleged	Proven
Sumatra	102	2,276	1,551	3,294	976	1,120
Java	75	1,259	1,191	3,403	377	1,820
Bali and Nusa Tenggara	40	295	169	996	231	42.5
Kalimantan	14	151	18	6	0	0
Sulawesi	91	1,365	343	1,063	180	120
Maluku	34	560	91	485	6	2
Papua	3	75	0	0	0	0
Total	**359**	**5,981**	**3,363**	**9,247**	**1,770**	**3,104.5**
		23,465.5				

Source: Compiledby the authors based on study [2].

Existing studies on social acceptance and multi-level governance (MLG) tend to treat acceptance as a local outcome or focus on inter-level coordination while neglecting power asymmetries. This paper addresses this gap by reconceptualizing social acceptance as a governance-mediated, politically contested process that emerges across governance levels. Using the Padarincang geothermal case, we demonstrate how vertically centralized governance produces a legitimacy gap between national ambitions and local acceptance, even under strong policy and market support. Our contribution is two fold: (1) we integrate MLG perspectives into social acceptance theory to explain cross-level misalignment, and (2) we show that project failure in the Global South often stems not from local opposition alone but from structural asymmetries in authority and participation.

2 Literature Review

2.1 Social Acceptance in Renewable Energy Development

The concept of social acceptance represents a key analytical framework in renewable energy studies. Wüstenhagen et al. [18] defined it as a three-dimensional process comprising socio-political acceptance (policy and public support), market acceptance (investor and consumer acceptance), and community acceptance (local response). This theory has since been refined for application in Global South contexts, where social dynamics are more complex [17, 19].

Social acceptance is not a static outcome but rather a dynamic process shaped by institutions, power relations, and historical experience [19, 20]. Consequently, resistance to renewable energy projects often reflects substantive concerns about environmental risks, distributive injustice, and procedural exclusion, rather than constituting irrational rejection. Empirical studies of geothermal energy across multiple countries (including Japan [21–23], Germany [24], Italy [25, 26], Switzerland [27, 28], the United States [29], Turkey [30, 31], East Africa [32], Kenya [33], and the Philippines [34]) reveal diverse patterns. Higher resistance is consistently observed among directly affected communities who express concerns about environmental degradation, water resources, and livelihoods [35, 36].

Macro-level explanations such as the 'Not In My Backyard' (NIMBY) syndrome frequently fail to address issues of local community rights and social justice [37]. In Indonesia and other Global South settings, social acceptance is largely determined by institutional trust, meaningful participation, and perceptions of distributive justice [32]. Wolsink emphasises that institutions must be placed at the core of analysis. He further argues that leaders are required to establish a fair and participatory institutional environment to foster acceptance [19, 20, 38].

Critically, Western frameworks of social acceptance are considered less capable of capturing realities in developing countries. In these contexts, historical marginalisation, weak governance, and power asymmetries fundamentally shape community responses [17, 19, 39]. Resistance, therefore, can be understood as a political expression of inequality. As such, the analysis of social acceptance must be situated within broader governance structures and political dynamics.

2.2 Multi-Level Governance and Energy Policy Implementation

MLG is a framework that conceptualizes policy-making and implementation as a process involving multiple levels of authority (local, national, and supranational) as well as governmental and non-governmental actors [30, 40, 41]. This framework operates beyond a purely top-down hierarchy. In energy policy, MLG enables the analysis of how national targets are translated to the local level, which frequently generates tensions between central priorities and local interests. Renewable energy development, including geothermal energy, exemplifies such tensions.

However, in many developing countries, MLG functions in an asymmetrical and hierarchical manner. The central government remains dominant, while local governments act primarily as implementers or mediators rather than as autonomous decision-makers [42]. This asymmetry is particularly evident in PSN, where central priorities tend to override local planning and community aspirations. Studies from Indonesia indicate a pattern of MLG characterized by fragmented authority, overlapping regulations, and limited participatory space for local actors [10]. Consequently, local governments face a structural dilemma: they must comply with central mandates while also responding to community pressures. This dual pressure results in weak coordination, legitimacy deficits, and social conflict.

Therefore, MLG should be understood not only as a coordination tool but also as a political arena where power and legitimacy are contested across governance levels.

2.3 Integrating Social Acceptance and Multi-Level Governance

Integrating the social acceptance framework with MLG is essential for understanding why renewable energy projects continue to face local resistance, even when they are supported by national policies and markets [19, 43]. From this integrated perspective, socio-political acceptance (constructed at the national level through policies and strategic narratives [18]) is often disconnected from community acceptance at the local level [44]. This disconnection arises because top-down and fragmented governance mechanisms reinforce mistrust and perceptions of distributive injustice. Governance dominated by a hierarchical approach limits meaningful participation and leads only to procedural acceptance. By contrast, intermediary actors (such as non-governmental organizations (NGOs) and social movements) transform local grievances into political pressure that influences policy dynamics at various levels [10]. Thus, resistance reflects not only local rejection but also failures of coordination and legitimacy within multilevel governance arrangements. This dynamic is particularly evident in the Global South, where national development agendas intersect with socio-ecological realities and unequal power relations.

3 Methodology

This study employed a qualitative research design with a case study approach [39]. This approach was appropriate for examining complex socio-political processes and actor interactions in depth. The case study approach enabled the research to capture contextual dynamics, power relations, and meaning-making processes surrounding the Padarincang geothermal project by integrating multiple sources of evidence [45].

The research site was Padarincang, Serang Regency. This site was selected because local communities had actively resisted the Kaldera Danau WKP since 2015, rendering it a critical case for analyzing social acceptance and MLG dynamics.

The study examined the perspectives of actors across multiple governance levels, including central, provincial, and local governments; political actors and local media; community leaders and residents; academics; and NGOs. These perspectives were analytically grouped not to homogenize actors but to reflect their functional positions within the governance system. The three functional groups were policy decision-makers (government actors), affected stakeholders (local communities), and intermediary or advocacy actors (academics and NGOs). This grouping enabled systematic comparison of how different governance roles shaped interpretations of the project and responses to it.

Data were collected through three primary methods. First, semi-structured interviews were conducted with stakeholders at national, regional, and local levels. Interviews with central and regional government officials focused on energy policy frameworks, regulatory instruments, and the rationale behind geothermal development. Interviews with community leaders and residents explored perceptions of risk, participation, trust, and impacts on livelihoods. Academics and NGO representatives were interviewed to capture their roles as intermediary actors, particularly in knowledge production, advocacy, and community mobilization related to geothermal development in Padarincang. The developer company, PT Sintesa Banten Geothermal (PT SBG), was not interviewed because it had been acquired and was no longer operational at the time of the study. Prior to interviews, informed consent was obtained from all participants. Participants were fully informed of the research objectives, the voluntary nature of participation, and their right to withdraw at any time. Interviews were conducted using interview guidelines and recorded with digital voice recorders.

Second, non-participant observations were conducted to document stakeholder interactions, public meetings, protest activities, and government–community engagements related to the project. Third, document analysis was undertaken, including national and regional policy documents such as RUEN, Regional Energy General Plan (Rencana Umum Energi Daerah, RUED), environmental permitting documents (Environmental Management Effort (Upaya Pengelolaan Lingkungan Hidup, UKL)/Environmental Monitoring Effort (Upaya Pemantauan Lingkungan Hidup, UPL) and Environmental Impact Assessment (Analisis Mengenai Dampak Lingkungan, AMDAL)), geothermal regulations, regional spatial plans (Rencana Tata Ruang Wilayah, RTRW), and other relevant official records.

Data analysis followed Braun and Clarke's thematic analysis framework [46]. This framework consisted of six iterative stages. First, interview transcripts, observational notes, and documents were transcribed and familiarized to identify recurring patterns. Second, initial codes were generated through open coding, capturing meaningful segments of data across actor groups. Third, codes were grouped into broader themes and sub-themes, including socio-political acceptance, market acceptance, community acceptance, intergovernmental coordination, cross-level policy conflict, and the implications of social acceptance for energy policy outcomes. Actor perspectives were compared within and across these themes to maintain analytical differentiation while enabling cross-actor interpretation. Fourth, themes were reviewed and refined to ensure coherence and alignment with the research questions. Fifth, themes were clearly defined and named to reflect their analytical scope. Sixth, the findings were interpreted in relation to theories of social acceptance and MLG and compared with existing studies to identify theoretical and practical contributions. NVivo 12 software was used to support data management and analysis, enabling systematic coding, transparent documentation of analytical decisions, and the organization of data hierarchies throughout the research process [47].

4 Results

4.1 Historical Trajectory of the Padarincang Geothermal Project

The historical trajectory of the conflict is summarized in Figure 1. The Padarincang geothermal project was formally initiated in 2009. The area was designated by the central government as the Kaldera Danau Banten WKP, and PT SBG was appointed as the developer. Subsequent regulatory reinforcement positioned Padarincang firmly within Indonesia's national renewable energy agenda. This reinforcement included the extension of exploration permits and the project's designation as a PSN and National Vital Object. During this early phase (2009–2014), the project progressed largely as a technocratic and regulatory process, with limited local engagement. Community responses remained relatively passive. This passive stance reflected restricted access to information, and residents perceived geothermal development as a distant, state-driven initiative rather than an imminent local intervention. Analytically, this phase illustrates how strong socio-political acceptance at the national level can coexist with weak local salience when governance remains centralized and socially insulated.

Figure 1. History of conflict geothermal in Pandarincang, Serang, Banten

As the project entered the exploration stage in 2015–2016, governance dynamics shifted from regulatory planning to material intervention. The entry of heavy machinery and the construction of road access transformed geothermal development from an abstract policy goal into a visible local disruption. These activities were carried out without meaningful prior consultation. At this stage, environmental permitting only required a UKL/UPL rather than a full AMDAL. However, the absence of transparent engagement amplified community concerns about forest conversion, water security, and agricultural livelihoods. Subsequent public outreach efforts failed to mitigate these concerns and instead deepened distrust. Communities perceived these efforts as procedural formalities rather than opportunities for deliberation. This phase demonstrates how procedural deficits within MLG can rapidly erode emerging community acceptance, particularly when decision-making authority remains concentrated at higher levels.

Open conflict emerged after 2017 and escalated through 2019, marked by organized resistance, mass mobilization, and the involvement of religious leaders, students, and local organizations. Despite the expiration of exploration permits, project activities continued, reinforcing perceptions of regulatory inconsistency and institutional bias. The framing of the project as a national vital object further intensified contestation. This framing securitized the conflict rather than addressing underlying governance grievances. By 2019, Padarincang had become a symbol of resistance against centralized energy development perceived as environmentally risky and socially unjust. Tensions briefly resurfaced in 2022. However, the project has remained inactive since 2018. Physical infrastructure has been abandoned, and there is no clear pathway for resumption before 2026–2027. Analytically, this trajectory illustrates how misalignment across governance levels can culminate in project paralysis despite strong formal regulatory support. This misalignment occurs between national legitimacy, subnational mediation, and local acceptance.

4.2 Social Acceptance in Geothermal Development in Banten Province

4.2.1 Socio-political acceptance

(1) Government support and the construction of national-level acceptance

Government support for geothermal development in Indonesia reflected a progressive consolidation of legal authority, market instruments, and institutional coordination. This consolidation positioned geothermal energy as a central pillar of the national energy transition.

Early regulatory arrangements under Law No. 27/2003 placed geothermal energy within the mining sector. This arrangement generated fragmented authority between central and regional governments, as well as regulatory uncertainty that constrained development. Law No. 21/2014 substantially addressed this fragmentation by reclassifying geothermal energy as a renewable resource and introducing the Geothermal Business License (Izin Usaha Pertambangan Panas Bumi, IUPB). These changes strengthened legal certainty, expanded investment opportunities, and permitted utilization in protected forest areas. Analytically, this shift demonstrated how socio-political acceptance was actively produced at the national level through legal redefinition and regulatory stabilization, rather than emerging from bottom-up consensus [48, 49].

This consolidation was reinforced through derivative regulations governing working area auctions, licensing, and electricity utilization. It was also reinforced through national planning instruments such as RUEN, which elevated geothermal energy as a priority in the national energy mix. Subsequent reforms under the Job Creation Law (Law No. 6/2023) further centralized authority by simplifying permits through a risk-based system and integrating environmental approvals into business licensing, particularly for PSN. Institutionally, the establishment of a dedicated Geothermal Directorate under the Directorate General of New, Renewable Energy and Energy Conservation (Energi Baru, Terbarukan, dan Konservasi Energi, EBTKE) strengthened cross-sectoral coordination among energy, environmental, planning, and financial agencies. From a MLG perspective, these arrangements enhanced policy coherence and investment confidence. However, they simultaneously narrowed the discretionary and deliberative space of subnational actors [50–52].

Beyond regulatory design, government support was operationalized through state-led risk mitigation and market stabilization mechanisms. Fiscal incentives, tariff guarantees through State-Owned Electricity Corporation (PT Perusahaan Listrik Negara (Persero), PT PLN), and access to international climate finance (such as Just Energy Transition Partnership (JETP) and multilateral development banks) reduced financial uncertainty. These instruments also positioned geothermal development as a low-risk, state-backed sector aligned with long-term decarbonization goals. In parallel, as shown in Figure 2, the Government Drilling scheme and the provision of geothermal potential data by the Geological Agency shifted early-stage exploration risks from private investors to the state. Theoretically, these instruments illustrated how market acceptance was constructed through policy insulation and public risk absorption. This construction reinforced national-level acceptance while decoupling investment decisions from local social conditions [53, 54].

At the subnational and local levels, support for the Padarincang geothermal project reflected strong vertical alignment with national priorities despite persistent social resistance. Provincial and district governments consistently framed the project as a PSN that was required to be supported on scientific and strategic grounds. This framing positioned local authorities primarily as facilitators of implementation rather than autonomous decision-makers. Outreach efforts involving security forces, academics, and village institutions sought to translate national legitimacy into local acceptance. However, these efforts were widely perceived as one-directional and procedurally closed. Village governments faced a structural dilemma, caught between compliance with central mandates and responsiveness to community opposition.

Analytically, this configuration revealed a key tension in MLG. Centralized support mechanisms successfully generated socio-political and market acceptance. However, community acceptance remained contingent on procedural legitimacy and meaningful deliberation. These could not be substituted by facilitation or securitization only [55–57].

Figure 2. Reverse and installed capacity of geothermal energy in Indonesia

Source: Adapted from Ref. [58] under the Creative Commons CC BY 4.0 license.

(2) Intermediary actors, knowledge mediation, and the contestation of legitimacy

Academic and civil society actors played a critical intermediary role in geothermal development. They mediated interactions between the state, the developers, and the local communities. Academic actors framed community resistance not merely as opposition to renewable energy. Instead, they interpreted it as a response to information asymmetries, procedural exclusion, and the neglect of cultural and socio-ecological considerations. Through public discourse, research dissemination, and policy engagement, academics emphasized three requirements: transparent environmental information, culturally sensitive engagement, and integration of geothermal development with local economic activities (such as agriculture, tourism, and education). Analytically, this role positioned academics as producers of epistemic legitimacy. They sought to bridge socio-political and community acceptance by reframing geothermal development as a socially embedded process rather than a purely technocratic one [59–61].

In addition to academic actors, civil society organizations (including Syarekat Perjuangan Rakyat (SAPAR), Wahana Lingkungan Hidup Indonesia (WALHI), Center of Economic and Law Studies (CELIOS), and Jaringan Advokasi Tambang (JATAM)) functioned as political and knowledge intermediaries. They translated localized grievances into broader policy-relevant claims. SAPAR, as a grassroots organization, mobilized local farmers, religious leaders, and youth. It articulated concerns about water sources, land use, and livelihood security. WALHI and JATAM expanded this resistance through national advocacy, legal framing, and rights-based narratives grounded in the principle of Free, Prior, and Informed Consent (FPIC). CELIOS contributed technical and scientific analyses that strengthened the evidentiary basis of community claims regarding environmental and seismic risks. Theoretically, these actors operated as cross-level translators: they converted everyday local concerns into discursive, legal, and scientific arguments that could enter national policy arenas. Consequently, they reshaped the contours of social acceptance beyond the community scale [62, 63].

While intermediary actors did not fundamentally alter central government decisions, their influence remained significant. This limitation was largely attributable to the project's embedding within the PSN and National Vital Object frameworks. At the local level, advocacy efforts delayed project activities, restricted site access, and prompted local governments to exercise greater caution when issuing derivative permits. At the national level, these actors successfully shifted the discourse on geothermal development from a narrowly technocratic narrative toward one that encompassed socio-ecological risks, water rights, and environmental justice. Analytically, this outcome demonstrated that intermediary actors might exert limited influence over formal policy decisions while substantially shaping discursive legitimacy and political salience within MLG systems [64].

This dynamic revealed a fundamental legitimacy gap in the Padarincang project. Despite the strong administrative and expert-based legitimacy conferred by the state, local communities rejected the project. Their grounds for rejection were clear: legality had been achieved without meaningful participation. Claims of minimal consultation, lack of access to AMDAL documentation, and the absence of FPIC underscored the distinction between legal legitimacy and social legitimacy. Theoretically, this finding reinforced a key argument: social acceptance cannot be secured through formal authorization or expert endorsement alone. Instead, it depends on procedural legitimacy, trust, and recognition of community rights. These dimensions are actively negotiated through intermediary actors across governance levels [65–67].

4.2.2 Market acceptance

From a financing perspective, the market acceptance dimension referred to the extent to which an energy project was economically viable and institutionally acceptable to key market actors, including developers, investors, off-takers, and regulators. In the Padarincang geothermal project, market acceptance was strongly constructed at the national and corporate levels through a combination of policy guarantees, fiscal incentives, and long-term contractual arrangements.

For PT SBG, geothermal energy was framed as a strategic asset within Indonesia's low-carbon transition. It offered predictable revenues through a Power Purchase Agreement (PPA) with PT PLN, low operational costs over a long project lifespan, and insulation from fossil fuel price volatility. Analytically, this finding reflected how market acceptance in energy transitions was not a spontaneous market outcome. Instead, it constituted an institutional product shaped by state intervention, risk socialization, and regulatory stabilization [68–70].

This market viability was further reinforced by the project's alignment with national renewable energy targets, its designation as a PSN, and its access to both domestic and international climate finance. Fiscal incentives, simplified permitting via the Online Single Submission system, and the prospect of multilateral funding collectively reduced investment risk and enhanced investor confidence. From a political economy perspective, these instruments demonstrated how the state actively constructed a favorable investment climate by transferring early-stage geological, regulatory, and market risks away from private actors. Theoretically, this confirmed that market acceptance operated through MLG arrangements, in which central governments played a dominant role in shaping investment rationalities and narrowing the space for market contestation [54].

However, the Padarincang case also revealed a critical contradiction within the concept of market acceptance. While the project enjoyed strong acceptance among investors and policymakers, it faced profound rejection from local communities. This rejection applied not only to the physical project but also to the electricity it would have produced. Residents explicitly rejected the consumption of geothermal electricity on ethical and political grounds. They framed such consumption as complicity in environmental harm and inter-regional injustice. This collective stance challenged the conventional assumption that consumers were passive end-users motivated primarily by price or reliability. Analytically, this finding extended the concept of market acceptance by demonstrating that consumer acceptance was socially and morally mediated. Specifically, it was shaped by solidarity, environmental ethics, and perceptions of distributive injustice, rather than by economic calculus alone [71].

Theoretically, this tension underscored a structural misalignment between nationally constructed market acceptance and locally grounded social acceptance [72, 73]. While state-backed market instruments succeeded in stabilizing investment expectations, they failed to internalize community-defined values related to environmental protection and moral legitimacy. This disjunction illustrated that market acceptance could not be treated as an autonomous dimension of social acceptance. Rather, it was contingent upon broader socio-political legitimacy and trust across governance levels. In the absence of such legitimacy, market acceptance became fragile and politically contested, even when financial and regulatory conditions appeared optimal.

4.2.3 Community acceptance

(1) Project benefits: between hope and fear

Local perceptions of the benefits of the Padarincang geothermal project were dominated by skepticism rather than optimism. The state and project developers framed geothermal development as a source of jobs, infrastructure improvement, and national energy security. However, these promises had not translated into tangible benefits at the community level. Employment for local residents remained largely temporary, low-skilled, and low-paid. Technical and long-term positions were perceived to be occupied by external workers. Analytically, this gap illustrated a classic distributive mismatch, where projected development gains were spatially and socially disconnected from those who bore the risks [74].

Infrastructure improvements, particularly road construction, were acknowledged by residents. However, these improvements were interpreted as serving project logistics rather than public needs. Basic services remained largely unchanged, including clean water, electricity, education, and healthcare. At the same time, agriculture (the community's primary livelihood) was perceived to be increasingly vulnerable.

This situation reflects a risk-benefit asymmetry that undermines distributive justice: local communities experience immediate and irreversible risks, while benefits remain abstract, future-oriented, and nationally framed [75, 76]. Such asymmetry is a widely recognized driver of resistance in renewable energy projects, especially in rural and resource-dependent communities.

A dominant narrative emerged from interviews and community forums. The project primarily benefited the state and corporate actors. Local residents were left with environmental disruption and social uncertainty. This perception was reinforced by unequal access to information and decision-making, where coordination with the developer and government was limited to a small group of local elites. Analytically, this elite-mediated participation transformed benefit-sharing into a symbol of exclusion, intensified distrust, and framed the project as extractive rather than developmental. Rather than fostering acceptance, promised benefits became a source of moral grievance that strengthened collective opposition.

(2) Environmental, social, and cultural risks

Environmental risks constituted the most salient basis for community resistance in Padarincang. Forests (leweng) were not perceived merely as ecological assets but as living infrastructures that sustained agriculture, water security, and social reproduction. Residents had already experienced floods, landslides, and agricultural damage following forest clearing and road construction. This experience reinforced fears of long-term ecological disruption.

Beyond local experience, community perceptions were shaped by collective learning from other geothermal sites in Indonesia. Examples included H_2S gas leaks in Sorik Marapi, water scarcity in Dieng, and technical accidents in Pangalengan. Advocacy networks and media circulated this information across regions. Analytically, this demonstrated how risk perception was socially constructed through translocal knowledge and historical memory, rather than being limited to site-specific impacts [77, 78].

Geothermal development was therefore framed not as "clean energy" but as part of a broader extractive trajectory associated with socio-ecological harm. Concerns about induced seismicity, thermal pollution, water depletion, and forest degradation reinforced a perception of irreversible environmental loss. From a theoretical standpoint, this finding aligned with scholarship on energy justice and socio-technical transitions. Low-carbon technologies can reproduce extractive dynamics when local ecological values are marginalized [79–81].

Socially and culturally, the project had generated horizontal tensions within the community. While a small minority viewed geothermal development as an economic opportunity, the majority interpreted support for the project as alignment with external interests. This division threatened long-standing norms of kinship, religious solidarity, and agrarian identity. The anticipated influx of outside workers further amplified fears of cultural erosion and moral disorder. Analytically, this showed that resistance was not merely instrumental but identity-based. Opposing the project became a means of defending cultural continuity and intergenerational responsibility [82]. In this sense, rejection of the geothermal project represented not opposition to development per se, but a collective effort to protect environmental integrity, social cohesion, and cultural meaning within the energy transition.

(3) Procedural exclusion, trust deficit, and the consolidation of collective resistance

In the Padarincang geothermal project, local communities perceived their involvement in project planning as minimal, procedural, and largely symbolic. Participation was limited to one-way socialization sessions with selected village elites. Most residents became aware of the project only after heavy equipment entered their area. Although the environmental permitting process formally followed existing regulations (initially requiring only a UKL/UPL during the exploration phase), the community viewed this regulatory design as structurally excluding meaningful participation. Analytically, procedural compliance coexisted with substantive exclusion, producing a legitimacy gap that undermined local social acceptance despite formal regulatory adherence.

This exclusion directly translated into a profound trust deficit toward both the developer (PT SBG) and government authorities at multiple levels. Residents had limited access to key documents (AMDAL, UKL/UPL, PPAs, and geothermal working area maps). Combined with delayed or selective information disclosure, this lack of access reinforced perceptions of opacity and manipulation. When residents attempted to question or contest the project, dissent was often politicized and framed as opposition to national development.

From a MLG perspective, this situation reflected a breakdown of vertical accountability. Centralized decision-making narrowed deliberative space at the local level. As a result, institutional credibility eroded, and governance coordination was transformed into a source of conflict [55].

The erosion of trust and procedural legitimacy became the foundation for sustained collective resistance. Opposition in Padarincang was not driven by demands for compensation or material bargaining. Instead, it was rooted in ecological, moral, and identity-based concerns. Springs, agricultural land, and mountains were understood as commons with social, cultural, and spiritual value. Defending them became a political act of safeguarding community sovereignty.

Religious leaders emerged as key mobilizing actors. They translated ecological risks into moral-religious narratives, framing environmental protection as an ethical obligation.

Consequently, the Padarincang case demonstrated that community resistance represented a political response to exclusionary governance, not merely a communication failure. This finding highlights trust and participation as key mediating mechanisms in social acceptance and MLG literature. When these mechanisms failed, local communities shifted from passive stakeholders to active political actors contesting state-led energy transitions [83–85].

4.3 Multi-Level Effect on Geothermal Energy Development in Indonesia

4.3.1 Fragmented vertical coordination in multi-level governance

The implementation of the Padarincang geothermal project revealed a pattern of weak and asymmetric coordination across government levels, a pattern characteristic of top-down governance in PSN. Decision-making authority was concentrated at the central level. The Ministry of Energy and Mineral Resources (MEMR), through the Directorate General of New, EBTKE, designated the project as a PSN and controlled the WKP tender in line with national energy targets. This designation effectively constrained provincial and district governments, which were positioned as implementers obligated to support the project rather than as autonomous decision-makers. Analytically, this configuration reflected a vertical and hierarchical model of MLG, in which policy coherence at the national level was achieved at the expense of deliberative capacity at lower levels [50, 86].

At the provincial and district levels, coordination largely took the form of mediation and socialization rather than substantive dialogue. Provincial authorities emphasized the strategic and low-carbon benefits of geothermal energy, while district governments facilitated communication between the developer and the community through formal forums involving bureaucratic, political, and security actors. However, these efforts remained narrowly procedural and securitized. The involvement of police and military forces to protect the project as a National Vital Object exemplified this securitization. From a governance perspective, such securitized coordination transformed socialization into compliance-seeking, reinforcing perceptions that the state prioritized project continuity over community concerns. Consequently, this dynamic undermined trust and socio-political acceptance [87].

The disjuncture between administrative coordination and communicative effectiveness became a central source of conflict. Although intergovernmental coordination appeared formally functional, it failed to reconcile the state's technocratic narrative of geothermal development with local interpretations of ecological, social, and cultural risk. Theoretically, this case demonstrated that vertical coordination alone was insufficient to secure social acceptance. When MLG lacked horizontal deliberation and downward accountability, it intensified legitimacy deficits and provoked organized resistance [86, 88]. The Padarincang case thus contributed to MLG literature by illustrating how national strategic framing stabilized policy authority while simultaneously destabilizing local legitimacy. This dual dynamic posed structural challenges to achieving a socially just energy transition.

4.3.2 Multi-level policy conflict and spatial governance tensions

Policy conflict in the Padarincang geothermal project reflected a structural tension between national strategic priorities, subnational regulatory constraints, and community claims over living space.

At the central level, the project was framed as part of the PSN. It was embedded within national commitments to accelerate renewable energy deployment and meet climate targets. This framing was reinforced through cross-agency coordination, in which economic, technical, legal, and security institutions participated. Analytically, this finding demonstrated how central governments constructed socio-political legitimacy through strategic narratives and institutional alignment, often at the expense of local regulatory considerations.

At the regional level, the Banten Provincial Government and the Serang Regency Government occupied an ambivalent position. They were formally obliged to support the project because of its PSN status. At the same time, they were constrained by the Serang Regency RTRW, which designated Padarincang as an agricultural, reforestation, and tourism area. This regulatory contradiction reduced local governments to intermediaries and administrative facilitators, depriving them of autonomous decision-making authority. From a MLG perspective, this demonstrated a misalignment between vertical authority and horizontal policy coherence. Overlapping legal frameworks generated implementation ambiguity and weakened local legitimacy [50, 88].

These tensions intensified further in the relationship between regional governments and the local community. Authorities often interpreted resistance as a product of misinformation or insufficient socialization. By contrast, residents framed the conflict as a violation of spatial justice and as the absence of a social license to operate. Community mobilization (through SAPAR, access blockades, and religious collective rituals) represented a counter-response. It opposed what residents perceived as the prioritization of national energy agendas over local ecological, agricultural, and spiritual values. Theoretically, this case highlighted how unresolved regulatory contradictions, particularly between national strategic policies and local spatial planning, could transform development projects into symbols of structural injustice, thereby catalyzing resistance and undermining social acceptance [72].

Overall, the Padarincang conflict underscored a core dilemma in energy transition governance: national decarbonization goals achieved formal authority and policy coherence at the center, yet generated legitimacy deficits at the periphery when they overrode local planning regimes. This finding contributed to MLG literature by showing that policy conflict was not merely administrative friction, but a governance mechanism through which power asymmetries, spatial justice, and community resistance were produced and contested.

4.3.3 Implications of low social acceptance for energy transition targets

Low social acceptance, particularly at the level of community acceptance, emerged as a critical bottleneck in the implementation of the Padarincang Geothermal Power Plant. More broadly, it also hindered Indonesia's ability to meet its national renewable energy targets. Although Banten Province possessed substantial renewable energy potential (including geothermal resources) the stalling of this nationally strategic project in Serang illustrated a key lesson: technical feasibility and economic viability alone were insufficient to ensure project realization. Analytically, this case demonstrated that targets became politically and institutionally unattainable, regardless of resource endowment [49, 72].

Community rejection in Padarincang was rooted in environmental, social, and cultural concerns. These concerns were linked to exclusionary planning processes, weak trust, and perceived threats to livelihoods and living space. These dynamics revealed a structural disjuncture between national energy transition narratives and local experiences of risk and injustice. From a theoretical perspective, this finding reinforced arguments within the social acceptance literature. That literature shows that community acceptance is not a derivative of socio-political or market acceptance. Instead, it is an autonomous dimension. When procedural and distributive justice are violated, community acceptance can veto project implementation [76, 89, 90].

The implications extended beyond the local and provincial levels. The failure to secure social acceptance in projects such as Padarincang contributed to explaining why Indonesia did not achieve its 2025 renewable energy mix target of 23%. This target was subsequently revised downward by the government, and the new deadline was postponed to 2030. Theoretically, this illustrated how micro-level governance failures could scale up into macro-level policy slippage. Community resistance therefore became directly linked to delays in national commitments under RUEN and international climate agreements. The Padarincang case underscored a critical lesson for energy transition governance: without embedding social legitimacy and participatory governance into project design, ambitious decarbonization targets risk remaining aspirational rather than achievable.

5 Discussion

5.1 Multi-Level Governance Interactions and the Shaping of Social Acceptance

The findings indicated that social acceptance of geothermal development was not generated at a single governance level. Instead, it emerged from interactions across national, subnational, and local governance arrangements. Acceptance did not evolve linearly from information provision or community attitudes alone. Rather, it was shaped by the distribution of authority, participation, and benefits across governance levels. Therefore, social acceptance should be conceptualized as a cross-level governance outcome, not a purely community-based phenomenon [18, 91].

At the national level, geothermal development received strong socio-political acceptance. This acceptance was supported by strategic energy policies, legal frameworks, and the project's designation as a PSN. Together, these factors constructed geothermal energy as a public good aligned with national development and climate objectives. However, the findings revealed limited downward translation of this legitimacy. Decision-making authority remained highly centralized, while local governments operated primarily as implementers rather than co-decision makers. Consequently, national-level socio-political acceptance did not automatically generate acceptance at subnational or community levels. This created a vertical legitimacy gap, which is consistent with critiques of hierarchical MLG in energy policy [48, 49, 92].

Similarly, market acceptance was reinforced through centralized policy instruments. These instruments included investment guarantees, licensing arrangements, and long-term power purchase agreements. They provided certainty for developers and investors [93, 94]. Although these mechanisms strengthened acceptance within market and policy circles, they remained weakly connected to community perceptions of benefit distribution. Governance arrangements prioritized national and corporate actors in determining compensation and development benefits. This prioritization limited community influence over distributive outcomes. This disconnect constrained community acceptance and reinforced perceptions of distributive injustice, even when opposition was not directed at geothermal technology itself [95, 96].

Community acceptance, meanwhile, was shaped primarily through procedural dynamics embedded in MLG. Although formal participation mechanisms existed (such as consultations and socialization programs) these processes were largely designed and controlled by actors operating at higher governance levels. This design positioned local governments and communities as recipients of information rather than as participants with decision-making authority. This form of procedural centralization weakened trust and limited the capacity of participation to foster acceptance. As previous studies suggested, symbolic or consultative participation detached from substantive influence may intensify contestation rather than mitigate it [97].

Importantly, the findings also highlighted the role of intermediary actors (such as NGOs and academics) in translating local grievances into broader political claims. These actors facilitated bottom-up cross-level interactions that reshaped governance debates beyond the local scale. This finding illustrates that community acceptance was not passively shaped from above. Instead, it could be actively renegotiated across levels in response to perceived exclusion [19, 61, 67]. Taken together, the analysis showed that acceptance outcomes were contingent on the alignment (or misalignment) of governance arrangements across levels, rather than on isolated local attitudes.

5.2 Implications for Global South and Resource-Dependent Governance Settings

While rooted in the Indonesian context, this study offers broader analytical insights into energy transition governance in resource-dependent and Global South settings. The Padarincang geothermal conflict reveals a recurring governance pattern in which renewable energy projects are advanced through centralized, technocratic, and security-oriented state logics, while social legitimacy and meaningful local participation are treated as secondary concerns. Consequently, renewable energy development (despite its green financing) often reproduces extractive governance practices inherited from fossil fuel-based trajectories, including top-down decision-making, weak social licensing, and coercive enforcement mechanisms [80, 98–100].

Theoretically, this study contributes to debates on MLG by demonstrating that the existence of multiple governance levels does not automatically lead to deliberative or inclusive outcomes. Instead, MLG operates in a vertical and disciplinary manner, wherein authority is centralized while responsibility for managing social conflict is displaced to subnational governments. This finding supports critical perspectives on MLG that emphasize power asymmetries, scalar dominance, and the prioritization of PSN over local governance autonomy [52, 101].

Moreover, the Padarincang case highlights that social acceptance must be understood as a political and moral process rather than merely as a function of information provision or economic compensation. Community resistance is rooted in ecological values, spiritual meanings, and claims over living space. This finding reinforces the argument that community acceptance constitutes a form of political agency and normative contestation, not a technical deficit to be corrected through improved communication [19, 49, 61, 102].

Finally, these findings contribute to discussions on energy transition justice in the Global South. They show how "green" transitions generate new forms of distributive and procedural injustice when embedded in authoritarian or developmentalist governance frameworks. By demonstrating that renewable energy projects can trigger legitimacy crises similar to those associated with fossil fuel extraction, this study warns against assuming that renewable energy is inherently socially benign. It also underscores the need for governance models that integrate social legitimacy, local autonomy, and ecological ethics as core elements of sustainable energy transitions [103–106].

6 Conclusions

Low community acceptance in the Padarincang geothermal project reflects a structured political response to exclusionary governance, not a lack of information or awareness. Despite strong national policy and market support, a persistent legitimacy gap separates formal project approval from social legitimacy at the community level.

This study advances a cross-level understanding of social acceptance by integrating social acceptance theory with MLG perspectives. While previous studies treat acceptance as a community-level attitude or a function of information and compensation, this article demonstrates that acceptance is a governance-mediated outcome shaped by interactions across national, subnational, and local institutions. The Padarincang case shows that centralized and vertical governance arrangements constrain participation, weaken trust, and generate legitimacy gaps, even when national-level socio-political and market acceptance is strong. Theoretically, this finding extends MLG scholarship by emphasizing power asymmetries, scalar dominance, and the disciplining role of PSN.

Empirically, the case illustrates that renewable energy development in resource-dependent contexts may reproduce extractive governance logics similar to fossil fuel projects. Community rejection (rooted in procedural exclusion, distributive injustice, and threats to ecological, cultural, and spiritual values) highlights that community acceptance constitutes a form of political agency. Local actors can renegotiate governance arrangements in response to exclusion, with intermediary actors (NGOs, academics, religious leaders) playing a crucial role in translating grievances into broader political claims.

The study also contributes to energy transition justice debates in the Global South. The misalignment between national decarbonization targets, spatial planning, and local livelihoods demonstrates that "green" transitions can generate new procedural and distributive injustices when embedded in centralized governance frameworks. Even when framed as environmentally beneficial, energy transitions may provoke social conflict and legitimacy crises if local autonomy, meaningful participation, and ecological ethics are marginalized.

Several limitations should be acknowledged. The study adopts a single-case design, which prioritizes analytical generalization over statistical breadth. While this approach provides nuanced insights into governance interactions, future cross-case and cross-technology comparisons would further illuminate how different governance configurations shape acceptance dynamics across the Global South.

Author Contributions

Conceptualizing and designing research, D.A.P., S.A., and I.R.W.; collected the data, D.A.P., S.A., I.R.W., and D.S.; conceived and designed the analysis, D.A.P. and M.S.; writing—original draft preparation and editing D.A.P., S.A., and I.R.W.; managing literature, D.A.P. and M.S.; supervision, D.A.P.; project administration, D.A.P. All authors have read and agree to the published version of the manuscript.

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Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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